

HTTP vs HTTPS

What is HTTP?

HTTP (HyperText Transfer Protocol) is a communication protocol used to transfer data between a web browser (client) and a web server. It allows users to access websites and exchange information over the internet.

Why HTTP is Used

- To establish communication between client and server.
- To request and receive web resources such as HTML pages, images, and files.
- To enable data exchange in web applications.

When HTTP is Used

- Accessing websites
- Downloading files
- Calling APIs
- Submitting forms

Limitation

HTTP transfers data in plain text, making it vulnerable to interception and security threats.

What is HTTPS?

HTTPS (HyperText Transfer Protocol Secure) is the secure version of HTTP. It uses **SSL/TLS encryption** to protect data transmitted between the client and server.

Why HTTPS is Used

- To secure user data during transmission.
- To prevent unauthorized access and data theft.
- To verify the authenticity of websites.
- To improve trust and website security.

When HTTPS is Used

- Online banking
- E-commerce websites
- Login and registration systems
- Payment gateway

Advantage

HTTPS encrypts data, making communication secure and reliable.

HTTP Request Methods

HTTP methods define how a client interacts with a server.

GET

Purpose

Retrieves data from the server.

Example

GET /users

Use Case

Viewing a list of users or fetching information.

POST

Purpose

Sends data to the server and creates a new resource.

Example

POST /users

Use Case

Creating a new account or submitting a form.

PUT

Purpose

Updates an existing resource completely.

Example

PUT /users/1

Use Case

Updating all details of a user profile.

PATCH

Purpose

Updates specific fields of an existing resource.

Example

PATCH /users/1

Use Case

Changing only a user's email or phone number.

DELETE

Purpose

Removes a resource from the server.

Example

DELETE /users/1

Use Case

Deleting a user account.

HEAD

Purpose

Retrieves only response headers without the response body.

Example

HEAD /users

Use Case

Checking if a resource exists before downloading.

DIAGRAM AS



HTTP AND HTTPS DOCUMENTATION



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- Payment gateways
- Secure web applications



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HTTP vs HTTPS

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Security	Not Secure	Secure
Encryption	No Encryption	SSL/TLS Encryption
Data Transfer	Plain Text	Encrypted Data
Port Number	80	443
Authentication	No	Yes
Browser Indicator	Not Secure ⚠️	🔒 Secure Lock
SEO Ranking	Lower	Higher
Recommended Use	Testing and Public Content	All Modern Websites

HTTP Request Methods

Method	Purpose	Example	Use Case	Data Sent
GET	Retrieves data from the server.	GET /users HTTP/1.1	Viewing a list of users or fetching information.	No (In URL)
POST	Sends data to the server and creates a new resource.	POST /users HTTP/1.1	Creating a new account or submitting a form.	Yes (In Body)
PUT	Updates an existing resource completely.	PUT /users/1 HTTP/1.1	Updating all details of a user profile.	Yes (In Body)
PATCH	Updates specific fields of an existing resource.	PATCH /users/1 HTTP/1.1	Changing only a user's email or phone number.	Yes (In Body)
DELETE	Removes a resource from the server.	DELETE /users/1 HTTP/1.1	Deleting a user account.	No (In URL)
HEAD	Retrieves only the response headers without the body.	HEAD /users HTTP/1.1	Checking if a resource exists before downloading.	No (Headers Only)
OPTIONS	Returns the list of supported HTTP methods for a resource.	OPTIONS /users HTTP/1.1	Checking available operations on an API endpoint.	No (In URL)

Difference Between HTTP Methods

Method	Purpose
GET	Retrieve Data
POST	Create Data
PUT	Update Entire Resource
PATCH	Update Partial Resource
DELETE	Remove Data
HEAD	Get Headers Only
OPTIONS	Show Supported Methods

Conclusion

HTTP is used for communication between a browser and a server, while HTTPS provides the same functionality with added security through encryption. For modern web applications, HTTPS is the preferred choice because it protects user data and ensures secure communication.

